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KALYANI GOVERNMENT ENGINEERING COLLEGE

(KGEC)

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FINAL RESEARCH REPORT — SUBMISSION COVER PAGE

PROJECT / RESEARCH TITLE

AI-Powered Flood Prediction, Impact Assessment & Emergency Response System

PROGRAMME DETAILS

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SUBMITTED BY

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UNDER THE MENTORSHIP OF

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DECLARATION

We hereby declare that this report is a record of genuine work carried out by us under the guidance of the above-mentioned mentor, as part of the IEEE SMC Student Branch Chapter, KGEC — Research Internship Programme 2026, and has not been submitted elsewhere for any other purpose.

Mentor's Signature

Programme Coordinator

IEEE SMC Student Branch Chapter, KGEC | Research Internship Programme 2026 | Date of Submission: *[DD / MM / 2026]*

Final Report of Internship Programme

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1. The report must be submitted as a single PDF file, named in the format: **IEEE-SMC-SBC-KGEC-[GroupNo].pdf**.
2. Use the official cover page template provided — do not alter the header, logo, or chapter branding.
3. Maximum report length: 8–10 pages (excluding cover page, references, and appendices), in A4 size, single-spaced, 11–12 pt font.
4. All students listed on the cover page must be actual contributors to the work; mentor approval is required before submission.
5. **AI-generated content must not exceed 20% of the total report text**, as measured by an approved AI-detection tool (e.g., Turnitin AI, GPTZero, or equivalent).
6. **Plagiarism (textual similarity) must not exceed 15% overall, with no single source contributing more than 3–5%**, as verified by a similarity-check tool.
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10. Citations and references must follow IEEE citation format throughout the report.
11. The declaration statement on the cover page must be signed by all students and the mentor before submission.
12. Reports must be submitted by the specified deadline; late submissions will only be accepted with prior written approval from the coordinator.
13. Each submission will undergo a mentor review followed by a chapter-level evaluation before final acceptance.
14. Any violation of the AI-usage or plagiarism thresholds will result in the report being returned for revision, or rejected in cases of repeated or severe violation.

General Instructions for Writing & Editing the Report

1. Write in clear, formal academic English — avoid contractions, slang, and overly casual phrasing.
2. Maintain a consistent tense throughout (past tense for work done, present tense for established facts/findings).
3. Structure the report logically: Introduction → Objective → Literature/Background → Methodology → Results & Discussion → Conclusion → References.
4. Each section should have a clear heading and follow a numbered format (e.g., 1. Introduction, 2. Methodology).
5. Keep paragraphs focused — one idea per paragraph, with clear topic sentences.
6. Use active voice wherever possible ("We implemented the model" rather than "The model was implemented by us").
7. Define all abbreviations and technical terms at first use before using the short form.
8. Number and caption all figures, tables, and equations, and refer to them explicitly in the text (e.g., "as shown in Fig. 2").
9. Proofread for grammar, spelling, and punctuation errors — use tools like Grammarly only for correction, not content generation.
10. Ensure consistent formatting: same font, heading style, spacing, and numbering throughout the document.
11. Avoid redundancy — do not repeat the same point across multiple sections.
12. Keep sentences concise; break long, complex sentences into shorter, readable ones.
13. Cross-check all numerical data, results, and claims for accuracy before final submission.
14. Have the mentor review at least one full draft before final submission, and incorporate feedback carefully.
15. Do a final read-through as a team to check flow, coherence, and alignment between all sections before converting

1. Introduction

Floods are among the most challenging natural hazards because they can develop rapidly and affect human settlements, infrastructure, transportation networks, and essential services. Effective disaster management therefore requires not only identifying the possibility of flooding but also estimating its potential impact and supporting timely emergency-response decisions. Conventional approaches that consider these aspects independently may provide limited support for coordinated decision-making during a flood event.

This project presents an **AI-Powered Flood Prediction, Impact Assessment & Emergency Response System** designed as an integrated disaster-intelligence pipeline. The system combines multiple artificial intelligence and data-driven components to provide a broader view of flood risk. The developed workflow includes flood forecasting, uncertainty estimation, satellite-image-based flood segmentation, impact assessment, risk fusion, and evacuation intelligence.

For flood forecasting, the project uses temporal environmental variables to model the relationship between rainfall, soil moisture, discharge, water level, and future flood conditions. A Bayesian neural-network approach is incorporated to provide both flood probabilities and an estimate of prediction uncertainty. This uncertainty information is subsequently used as part of the overall risk assessment rather than relying only on a single deterministic prediction.

The system also incorporates satellite-image-based flood segmentation using a U-Net architecture. The segmentation component identifies flooded regions in image data and provides an estimate of the flooded-area fraction. The resulting information is combined with impact-related factors such as affected population and infrastructure characteristics to estimate the potential consequences of a flood scenario.

To support emergency response, the project performs risk fusion by combining flood probability, prediction uncertainty, flooded area, and impact-related information into a unified risk assessment. The resulting risk information is further used to generate evacuation priorities and recommend suitable shelters. A FastAPI-based backend and Streamlit dashboard are included to present risk scenarios, Bayesian forecasting statistics, flooded-area information, and evacuation recommendations in an integrated interface.

The overall objective of the system is therefore to connect **prediction, assessment, risk analysis, and response** within a single AI-assisted framework. Rather than treating flood forecasting as an isolated prediction problem, the project demonstrates how different analytical components can be combined to support disaster-management and emergency-response planning.

2. Objectives

The primary objective of the proposed **AI-Powered Flood Prediction, Impact Assessment & Emergency Response System** is to develop an integrated framework that combines flood prediction, spatial assessment, risk analysis, and emergency-response support. The specific objectives of the project are:

1. **To develop an AI-based flood forecasting component** capable of estimating the likelihood of a flood event using relevant environmental and hydrological parameters.
2. **To incorporate uncertainty estimation into flood prediction** using a Bayesian neural-network approach, allowing the system to represent both predicted flood probability and associated uncertainty.
3. **To identify and estimate flooded regions from satellite imagery** using a U-Net-based image segmentation approach.

4. **To assess the potential impact of flooding** by considering spatial and infrastructure-related factors such as population, buildings, roads, hospitals, schools, flood depth, and flood duration.
5. **To develop an integrated risk-fusion mechanism** that combines flood probability, prediction uncertainty, flooded area, and impact-related information to produce an overall flood-risk assessment.
6. **To support emergency evacuation planning** by analysing evacuation nodes and road-network information and generating evacuation priorities and shelter recommendations.
7. **To provide an integrated visualization and decision-support interface** through which flood forecasts, risk information, impact assessment, and evacuation recommendations can be accessed.
8. **To demonstrate an end-to-end disaster-management workflow** connecting prediction, flood mapping, impact assessment, risk classification, and emergency response within a unified system.

3. Literature Review / Background

Flood prediction and disaster management have traditionally involved the analysis of hydrological, meteorological, and geographical information to identify areas that may be affected by flooding. With the increasing availability of environmental data and advances in artificial intelligence, machine-learning and deep-learning techniques have been increasingly incorporated into different stages of the flood-management process.

3.1 Flood Prediction Using Machine Learning

Flood forecasting involves analysing variables such as rainfall, soil moisture, river discharge, water level, and other environmental conditions to estimate the likelihood of future flooding. Machine-learning models can learn relationships between these variables and flood conditions from historical or generated observations. Neural-network-based approaches are particularly useful for modelling nonlinear relationships between environmental variables.

In the proposed system, flood forecasting is treated as one component of a larger disaster-management workflow. Instead of using the prediction alone, the resulting flood probability is subsequently incorporated into the risk-fusion stage.

3.2 Uncertainty-Aware Flood Prediction

A limitation of conventional predictive models is that a single prediction does not necessarily indicate how confident the model is in that prediction. In disaster-management applications, uncertainty can be important because decisions may need to account for situations in which the available information is incomplete or the prediction is less reliable.

Bayesian neural networks provide a mechanism for incorporating uncertainty into neural-network predictions. In the proposed system, the Bayesian forecasting component generates flood probabilities together with uncertainty-related information. This information is later used as an input to the overall risk-assessment process.

3.3 Satellite Image Segmentation for Flood Mapping

Remote-sensing imagery provides a means of identifying spatial changes associated with flooding over large areas. Image-segmentation techniques can be used to classify individual pixels or regions according to whether they represent flooded or non-flooded areas.

U-Net is a convolutional neural-network architecture widely used for image segmentation because it combines feature extraction with spatial information from different levels of the network. The proposed project uses a U-Net-based segmentation component to identify flooded regions in satellite-image data and estimate the proportion of an image affected by flooding.

3.4 Flood Impact Assessment

Flood prediction alone does not indicate the extent of potential damage. Two locations with similar flood probabilities may experience significantly different consequences depending on their population, buildings, transportation infrastructure, critical facilities, flood depth, and flood duration.

The proposed system therefore includes a separate impact-assessment component. Spatial information related to population, buildings, roads, hospitals, schools, vulnerability, flood depth, and flood duration is considered to estimate potential flood impacts.

3.5 Risk Fusion and Emergency Response

An effective disaster-management system needs to combine information from different sources rather than considering each prediction independently. Risk-fusion approaches can integrate hazard probability with uncertainty, spatial flood extent, and potential impact to produce a more comprehensive representation of risk.

The proposed system extends this concept by connecting risk assessment with emergency-response planning. Evacuation nodes and road-network information are incorporated to generate evacuation priorities and shelter recommendations. This creates a connection between **hazard prediction, impact assessment, risk classification, and emergency response**.

3.6 Background and Motivation for the Proposed System

The major motivation for the proposed system is the need to integrate multiple stages of flood disaster management into a single analytical workflow. Flood forecasting provides information about the likelihood of an event, satellite segmentation provides information about affected areas, and impact assessment provides information about potential consequences. Combining these outputs can provide a more comprehensive basis for risk assessment and emergency planning.

Accordingly, the proposed system brings together **Bayesian flood forecasting, satellite-image segmentation, flood-impact assessment, risk fusion, and evacuation intelligence** within one framework. This integrated approach forms the foundation for the methodology presented in the following section.

4. Methodology

The proposed system follows a multi-stage methodology in which flood forecasting, flood segmentation, impact assessment, risk fusion, and evacuation intelligence are connected to form an integrated disaster-management workflow. The overall process is organized into the following stages.

4.1 System Workflow

The methodology begins with environmental and hydrological data for flood forecasting. The forecasting component estimates the probability of a future flood event and provides uncertainty information using a Bayesian neural-network model.

In parallel, satellite-image data are processed using a U-Net-based segmentation model to identify flooded regions. The predicted segmentation is then used to estimate the proportion of an image affected by flooding.

The outputs from the forecasting and segmentation components are combined with spatial impact information. The impact-assessment component considers factors such as population, buildings, roads, hospitals, schools, vulnerability, flood depth, and flood duration.

The resulting information is passed to a risk-fusion stage, where flood probability, prediction uncertainty, flooded area, and impact-related variables are combined to produce an overall risk assessment. Finally, evacuation-node and road-network information is used to support evacuation prioritization and shelter recommendations.

Overall workflow:

Environmental Data → Flood Forecasting → Probability & Uncertainty
Satellite Images → U-Net Segmentation → Flooded Area
Impact Data → Impact Assessment
Probability + Uncertainty + Flooded Area + Impact → Risk Fusion
Risk Assessment + Evacuation Network → Evacuation Recommendations

4.2 Data Preparation

The project uses multiple datasets corresponding to different stages of the disaster-management pipeline. The flood-forecasting data contain environmental and hydrological variables used to predict future flood conditions. The forecasting dataset contains hourly observations and includes variables such as rainfall, soil moisture, discharge, water level, and other relevant environmental parameters.

For satellite-image segmentation, the project uses image data together with corresponding flood masks. The dataset is divided into training, validation, and testing subsets so that the segmentation model can be trained and evaluated independently.

The impact-assessment component uses spatial records containing information related to population, buildings, roads, critical facilities, vulnerability, flood depth, and flood duration. Evacuation data include locations and road-network information required for emergency-response analysis.

4.3 Flood Forecasting Using Bayesian Neural Network

The flood-forecasting component uses a neural-network-based model to estimate the probability of flooding during the future prediction period. The input variables represent environmental and hydrological conditions, while the target variable represents the occurrence of a future flood condition.

A Bayesian neural-network approach is used to incorporate uncertainty into the prediction process. Instead of producing only a deterministic output, the model provides a flood probability together with uncertainty-related information. These outputs are retained for use in the subsequent risk-fusion stage.

The forecasting data are separated into training and testing subsets. The neural network is trained using the available training observations, while the test data are used to obtain predictions for evaluation and downstream analysis.

4.4 Satellite Flood Segmentation Using U-Net

The satellite-image component uses a U-Net architecture for pixel-level flood segmentation. Each input image is processed by the network to generate a corresponding segmentation mask indicating the predicted flooded region.

The U-Net architecture contains an encoder component for extracting image features and a decoder component for reconstructing spatial information. Skip connections between corresponding encoder and decoder stages help preserve spatial details required for accurate segmentation.

The model is trained using the training subset and evaluated using validation and test data. Segmentation performance is assessed using metrics including **Dice coefficient** and **Intersection over Union (IoU)**. The predicted mask is also used to estimate the percentage of the image covered by flooding.

4.5 Flood Impact Assessment

The impact-assessment stage evaluates the potential consequences associated with a flood scenario. Spatial attributes such as population, buildings, roads, hospitals, schools, vulnerability, flood depth, and flood duration are considered.

These variables allow the system to move beyond identifying whether flooding may occur and instead consider **what may be affected**. The resulting impact information becomes an input to the risk-fusion stage.

4.6 Risk Fusion

Risk fusion integrates outputs from the different analytical components into a unified flood-risk representation. The main information sources include:

- Flood probability from the forecasting model
- Prediction uncertainty from the Bayesian model
- Flooded-area information from satellite segmentation
- Impact-related information from spatial assessment

Combining these factors provides a broader representation of flood risk than relying on flood probability alone. The resulting risk information is used to classify and prioritize scenarios according to their estimated level of risk.

4.7 Evacuation Intelligence

The emergency-response component uses evacuation nodes and road-network information to support evacuation planning. Based on the resulting risk assessment, locations can be prioritized according to the severity of the potential flood impact.

The system also associates evacuation locations with recommended shelters, providing information that can support emergency-response planning. This component connects the analytical risk-assessment stage with an actionable emergency-response output.

4.8 System Integration and Visualization

The individual components are integrated through a software-based interface. A **FastAPI backend** provides the application-level services, while a **Streamlit dashboard** provides a visualization and interaction layer.

The dashboard brings together information from the forecasting, segmentation, impact-assessment, risk-fusion, and evacuation components. This allows the outputs of the different stages to be viewed within a unified disaster-management interface rather than as isolated model results.

5. Results and Discussion

The developed system produces outputs at different stages of the flood-management workflow, including flood forecasting, satellite-based flood segmentation, flood-impact assessment, risk fusion, and evacuation planning. The results demonstrate how these components can be connected to provide an integrated view of flood risk.

5.1 Flood Forecasting Results

The flood-forecasting component was trained using the available environmental and hydrological observations. The model generates a probability associated with the occurrence of a future flood condition along with uncertainty information obtained through the Bayesian modelling approach.

The forecasting stage produced **43,860 test predictions**, which were subsequently made available for downstream risk analysis. The probability and uncertainty outputs provide additional information that can be incorporated into the risk-fusion stage.

The project does **not provide a sufficiently supported conventional classification accuracy, precision, recall, or F1-score for the Bayesian forecasting component**. Therefore, these metrics are not reported here to avoid making unsupported claims.

5.2 Satellite Flood Segmentation Results

The U-Net-based segmentation component was evaluated on the test dataset. The model achieved the following reported results:

Metric	Test Result
Dice Coefficient	**0.999786**
Intersection over Union (IoU)	**0.999572**
Flooded-Area MAE	**0.0046 percentage points**

The Dice coefficient and IoU indicate a high degree of overlap between the predicted flood segmentation masks and the corresponding test masks. The flooded-area estimation also produced a low mean absolute error according to the project's recorded evaluation results.

These results indicate that the developed segmentation component was able to identify the flooded regions in the test image data with high overlap relative to the corresponding reference masks.

5.3 Flood Impact Assessment

The impact-assessment component processes spatial information associated with flood scenarios. The project contains **12,000 spatial cells** with attributes related to population, buildings, roads, hospitals, schools, vulnerability, flood depth, flood duration, and estimated damage.

The inclusion of these variables allows the system to represent potential consequences in addition to the predicted occurrence of flooding. This is important because the severity of a flood event cannot be represented by flood probability alone. Areas with greater population density or critical infrastructure may require higher response priority even when their flood probability is comparable to other locations.

5.4 Risk-Fusion Results

The risk-fusion stage combines information produced by the individual components of the system. The project contains **200 risk scenarios**, incorporating flood probability, prediction uncertainty, flooded-area information, and impact-related information.

This stage provides a unified risk representation by combining hazard-related and consequence-related information. The resulting risk assessment can therefore be used as an intermediate decision-support layer between model predictions and emergency-response planning.

5.5 Evacuation and Emergency-Response Results

The project includes evacuation-network information consisting of **400 evacuation nodes and 760 road edges** in the corresponding network data. The processed evacuation results contain **253 locations** with information related to evacuation priority and recommended shelter.

These outputs demonstrate the connection between the risk-assessment stage and emergency-response planning. Instead of stopping at the identification of flood risk, the system produces location-level information that can assist in prioritizing evacuation and identifying suitable shelter options.

5.6 System Dashboard

The different outputs are integrated into the DisasterGuard AI application through a FastAPI backend and Streamlit-based dashboard. The dashboard provides access to information related to flood risk, Bayesian forecasting statistics, satellite segmentation, impact assessment, and evacuation intelligence.

The integrated interface demonstrates the project's overall workflow: environmental information is processed to generate flood predictions, satellite data are used to identify flooded areas, impact information is incorporated into risk assessment, and the resulting risk information is used to support evacuation decisions.

5.7 Discussion

The results demonstrate the feasibility of combining multiple AI and data-processing techniques within a unified flood-management framework. The strongest quantitative results are available for the satellite segmentation component, where the recorded test Dice coefficient and IoU are both close to 1.0.

The Bayesian forecasting component contributes a different type of information by providing probability and uncertainty rather than only a binary flood decision. The impact-assessment and risk-fusion components then extend the analysis from prediction to potential consequences. Finally, the evacuation component converts risk information into response-oriented outputs.

However, the results should be interpreted within the scope of the available project data. The forecasting data and evaluation setup should not be presented as evidence of operational performance in real-world flood events unless additional real-world validation is available. Similarly, the reported segmentation performance applies to the project's test dataset and should not automatically be generalized to all satellite imagery or geographic conditions.

Overall, the results support the project's central concept of integrating **flood prediction, flood mapping, impact assessment, risk analysis, and emergency response** into a single AI-assisted disaster-management workflow.

6. Conclusion

This project presented an **AI-Powered Flood Prediction, Impact Assessment & Emergency Response System** that integrates multiple stages of flood disaster management into a unified framework. The developed system combines flood forecasting, uncertainty estimation, satellite-image-based flood segmentation, flood-impact assessment, risk fusion, and evacuation intelligence.

The Bayesian neural-network component provides flood probabilities together with uncertainty information, allowing the forecasting results to contribute to a broader risk assessment rather than being treated as an isolated prediction. The U-Net-based satellite segmentation component provides spatial identification of flooded regions and achieved a test Dice coefficient of **0.999786** and an Intersection over Union of **0.999572** on the project's evaluation data. Flooded-area estimation also achieved a recorded mean absolute error of **0.0046 percentage points**.

The impact-assessment component extends the analysis by incorporating population, infrastructure, critical facilities, vulnerability, flood depth, and flood duration. These outputs are combined with forecasting and segmentation information through the risk-fusion stage to provide a more comprehensive representation of potential flood risk. The evacuation component further connects risk assessment with emergency response by generating evacuation priorities and shelter recommendations.

The integration of these components through the DisasterGuard AI application demonstrates the potential of combining artificial intelligence and spatial data analysis for disaster-management decision support. The system provides an end-to-end workflow from **flood prediction to impact assessment and emergency-response planning**.

At the same time, the reported results should be considered within the scope of the available datasets and evaluation setup. Further validation using larger, diverse, and real-world flood-event datasets would be required before applying the system as an operational disaster-response solution. Future development could also focus on real-time data integration, broader geographic validation, improved evacuation-route optimization, and deployment with continuously updated environmental and satellite observations.

Overall, the project demonstrates that integrating prediction, uncertainty estimation, spatial flood mapping, impact assessment, risk fusion, and evacuation intelligence can provide a stronger foundation for AI-assisted flood disaster-management systems than considering any single component independently.